

Top 20 Java Streams Coding Interview Questions with Code Answers

1. Find the maximum number in a list

```
List<Integer> numbers = List.of(5, 3, 9, 1, 7);  
int max = numbers.stream().max(Integer::compare).orElseThrow();  
System.out.println(max); // 9
```

2. Count even numbers in a list

```
long count = numbers.stream().filter(n -> n % 2 == 0).count();  
System.out.println(count);
```

3. Convert list of strings to uppercase

```
List<String> names = List.of("john", "doe");  
List<String> upper = names.stream().map(String::toUpperCase).toList();  
System.out.println(upper); // [JOHN, DOE]
```

4. Filter names starting with 'A'

```
List<String> filtered = names.stream().filter(n ->  
n.startsWith("A")).toList();  
System.out.println(filtered);
```

5. Find duplicate elements in a list

```
List<Integer> list = List.of(1, 2, 2, 3, 4, 4, 5);
Set<Integer> seen = new HashSet<>();
List<Integer> duplicates = list.stream()
    .filter(n -> !seen.add(n))
    .toList();
System.out.println(duplicates); // [2, 4]
```

6. Get first 3 elements of a list

```
List<Integer> firstThree = numbers.stream().limit(3).toList();
System.out.println(firstThree);
```

7. Skip first 2 elements

```
List<Integer> skipped = numbers.stream().skip(2).toList();
System.out.println(skipped);
```

8. Sum of all numbers in list

```
int sum = numbers.stream().mapToInt(Integer::intValue).sum();
System.out.println(sum);
```

9. Check if all numbers are even

```
boolean allEven = numbers.stream().allMatch(n -> n % 2 == 0);
System.out.println(allEven);
```

10. Check if any number is divisible by 5

```
boolean anyDivBy5 = numbers.stream().anyMatch(n -> n % 5 == 0);  
System.out.println(anyDivBy5);
```

11. Sort list in descending order

```
List<Integer> sortedDesc = numbers.stream()  
    .sorted(Comparator.reverseOrder())  
    .toList();  
System.out.println(sortedDesc);
```

12. Remove duplicates from list

```
List<Integer> unique = numbers.stream().distinct().toList();  
System.out.println(unique);
```

13. Find average of a list of numbers

```
double avg = numbers.stream().mapToInt(i -> i).average().orElse(0.0);  
System.out.println(avg);
```

14. Group strings by length

```
Map<Integer, List<String>> grouped = names.stream()  
    .collect(Collectors.groupingBy(String::length));  
System.out.println(grouped);
```

15. Join all strings with comma

```
String joined = names.stream().collect(Collectors.joining(", "));
System.out.println(joined);
```

16. Partition list into even and odd

```
Map<Boolean, List<Integer>> partitioned = numbers.stream()
    .collect(Collectors.partitioningBy(n -> n % 2 == 0));
System.out.println(partitioned);
```

17. Find second highest number

```
int secondMax = numbers.stream()
    .sorted(Comparator.reverseOrder())
    .distinct()
    .skip(1)
    .findFirst()
    .orElseThrow();
System.out.println(secondMax);
```

18. FlatMap list of lists

```
List<List<String>> listOfLists = List.of(List.of("a", "b"), List.of("c",
"d"));
List<String> flat = listOfLists.stream()
    .flatMap(List::stream)
    .toList();
System.out.println(flat); // [a, b, c, d]
```

19. Find frequency of each character in a string

```
String input = "stream";
Map<Character, Long> freq = input.chars()
    .mapToObj(c -> (char) c)
    .collect(Collectors.groupingBy(c -> c, Collectors.counting()));
System.out.println(freq);
```

20. Get top 3 highest numbers

```
List<Integer> top3 = numbers.stream()
    .sorted(Comparator.reverseOrder())
    .limit(3)
    .toList();
System.out.println(top3);
```
